

附录一：典型相关性分析的相关系数矩阵

组一：相关系数矩阵  $R_{11}$

|     | x1      | x2      | x3      | x4      | x5      | x6      | x7      | x8      | x9      | x10     | x11     |
|-----|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| x1  | 1.00    | 0.9262  | -0.7751 | 0.1026  | -0.8152 | -0.7132 | -0.0461 | 0.1619  | 0.2323  | -0.3775 | 0.0066  |
| x2  | 0.9262  | 1.0000  | -0.6109 | 0.1364  | -0.6627 | -0.5708 | 0.0240  | 0.0915  | 0.1842  | -0.2856 | 0.0160  |
| x3  | -0.7751 | -0.6109 | 1.0000  | -0.1511 | 0.8910  | 0.8407  | 0.2705  | -0.0686 | -0.0392 | 0.4677  | 0.0514  |
| x4  | 0.1026  | 0.1364  | -0.1511 | 1.0000  | -0.1344 | -0.4028 | 0.1942  | 0.0456  | -0.0529 | -0.1540 | -0.2247 |
| x5  | -0.8152 | -0.6627 | 0.8910  | -0.1344 | 1.0000  | 0.9288  | 0.1809  | 0.0131  | 0.0371  | 0.4797  | 0.0036  |
| x6  | -0.7132 | -0.5708 | 0.8407  | -0.4028 | 0.9288  | 1.0000  | 0.0765  | 0.0333  | 0.0808  | 0.4866  | 0.0494  |
| x7  | -0.0461 | 0.0240  | 0.2705  | 0.1942  | 0.1809  | 0.0765  | 1.0000  | 0.3170  | 0.1668  | 0.2166  | 0.1989  |
| x8  | 0.1619  | 0.0915  | -0.0686 | 0.0456  | 0.0131  | 0.0333  | 0.3170  | 1.0000  | 0.7116  | 0.1471  | -0.0058 |
| x9  | 0.2323  | 0.1842  | -0.0392 | -0.0529 | 0.0371  | 0.0808  | 0.1668  | 0.7116  | 1.0000  | 0.2265  | 0.0030  |
| x10 | -0.3775 | -0.2856 | 0.4677  | -0.1540 | 0.4797  | 0.4866  | 0.2166  | 0.1471  | 0.2265  | 1.0000  | 0.0777  |
| x11 | 0.0066  | 0.0160  | 0.0514  | -0.2247 | 0.0036  | 0.0494  | 0.1989  | -0.0058 | 0.0030  | 0.0777  | 1.0000  |

组二：相关系数矩阵  $R_{12}$

|     | y1      | y2      | y3      | y4      |
|-----|---------|---------|---------|---------|
| x1  | 0.0468  | 0.1221  | 0.3152  | -0.0548 |
| x2  | -0.0085 | 0.0562  | 0.2781  | 0.0920  |
| x3  | -0.0671 | -0.0351 | -0.1267 | 0.1649  |
| x4  | -0.2241 | -0.0103 | 0.0858  | 0.1120  |
| x5  | -0.0533 | -0.1358 | -0.1410 | 0.1003  |
| x6  | 0.0255  | -0.1112 | -0.1286 | 0.0781  |
| x7  | -0.1584 | -0.1137 | 0.0010  | 0.0997  |
| x8  | -0.1094 | 0.0089  | 0.2182  | 0.1142  |
| x9  | -0.0547 | 0.0558  | 0.1799  | 0.0569  |
| x10 | 0.0941  | -0.2668 | -0.1216 | 0.1999  |
| x11 | -0.1142 | -0.0488 | 0.0420  | 0.1098  |

组三：相关系数矩阵  $R_{22}$

| y1 | y2 | y3 | y4 |
|----|----|----|----|
|----|----|----|----|

```

y1  1.0000 0.0863 0.1492 0.0051
y2  0.0863 1.0000 0.3785 -0.0046
y3  0.1492 0.3785 1.0000 0.0601
y4  0.0051 -0.0046 0.0601 1.0000

```

附录二：

#### 1 无量纲处理程序

```

function data=get_unit(data0)
data0_max=max(data0)
data0_min=min(data0)
d=data0_max-data0_min
data0_size=size(data0);
for i=1:data0_size(1)
    for j=1:data0_size(2)
        data(i,j)=(data0(i,j)-data0_min(j))/d(j);
    end
end

```

#### 2 生成相关矩阵程序

```

function r_maxtri=relation_maxtri(data1,data2)
data1_size=size(data1)
data2_size=size(data2)
r_maxtri=zeros(data1_size(1),data2_size(1));
for i=1:data1_size(1)
    for j=1:data2_size(1)
        r12=cov(data1(i,:),data2(j,:))
        r_maxtri(i,j)=r12(1,2);
    end
end

```

#### 3 典型相关性检验程序

```

function prove(r11,r12,r21,r22,n,a)
r=get_mixr(r11,r12,r21,r22);
A=inv(r11)*r12*inv(r22)*r21;
B=inv(r22)*r21*inv(r11)*r12;
A_size=size(A);
B_size=size(B);
p1=A_size(1);
p2=B_size(1);
[la,va]=eig(A)
[lb,vb]=eig(B)
m=n-1-0.58*(p1+p2+1);
if p1<p2
    C=A;p=p1
else
    C=B;p=p2
end

```

```

[lc,vc]=eig(C);
v=zeros(p,1)
for i=1:p
    v=v+vc(:,i);
end
v=v';
v_card=zeros(1,p);
for i=1:p
    for j=1:p
        if v(i)<=v(j)
            v_card(i)=v_card(i)+1;
        end
    end
end
v_card
for i=1:p
    laa(:,v_card(i))=la(:,i);
    lbb(:,v_card(i))=lb(:,i);
    vcc(:,v_card(i))=vc(:,i);
end
laa
lbb
vcc
v(:)=0;
for i=1:p
    v=v+vcc(i,:);
end
v
Vh(1:p)=1;
Q(1:p)=0;
Xa(1:p)=0;
for i=1:p
    for j=i:p
        Vh(i)=Vh(i)*(1-v(j));
    end
    Q(i)=-m*log(Vh(i));
    e=(p1-i+1)*(p2-i+1);
    Xa(i)=chi2inv(1-a,e);
end
v
v.^0.5
Vh
Q
Xa

```

#### 4 秩和检验程序

```
function rr=zhjy(x0,y0)
x=cut0(x0)
y=cut0(y0)
x_size=size(x);
y_size=size(y);
if x_size(2)>y_size(2)
    b=x;
    x=y;
    y=b;
    x_size=size(x);
    y_size=size(y);
end
z=zeros(2,x_size(2)+y_size(2));
z(2,1:x_size(2))=1;
z(2,x_size(2)+1:x_size(2)+y_size(2))=2;
z(1,1:x_size(2))=x;
z(1,x_size(2)+1:x_size(2)+y_size(2))=y;
z
w=zeros(2,1);
for i=1:x_size(2)+y_size(2)
    for j=i+1:x_size(2)+y_size(2)
        if z(1,i)>=z(1,j)
            w=z(:,i);
            z(:,i)=z(:,j);
            z(:,j)=w;
        end
    end
end
z
r1=0;
for i=1:x_size(2)+y_size(2)
    if z(2,i)==1
        i
        r1=r1+i
    end
end
n1=x_size(2);
n2=y_size(2);
ER1=n1*(n1+n2+1)/2;
DR1=n1*n2*(n1+n2+1)/12;
rr=abs(r1-ER1)/(DR1^0.5)
%Za=norminv(1-a,0,1)
```